Section 04

Industrial Security (Bachelor)

IS Security

Section 04 – Industrial Protocols

Industrial Protocols

Lecture Notes

Department of Computer Science

Outline – Section 04

Learning Objectives

By the end of this section you will be able to

- Recognise common industrial protocols and their use cases.
- Understand why legacy protocols typically lack authentication.
- Decode a Modbus TCP frame by hand.
- Compare insecure legacy protocols with secure modern alternatives (OPC UA, MQTT-TLS).
- Identify when secure variants are appropriate.

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Legacy Heritage

Why Industrial Protocols Are Different

Core fact

Most industrial protocols in production today were **designed before the Internet** and provide no authentication, integrity, or confidentiality.

Designed 1970s–90s for serial links → Optimised: small payloads, low CPU, deterministic → Missing: authentication, encryption, integrity

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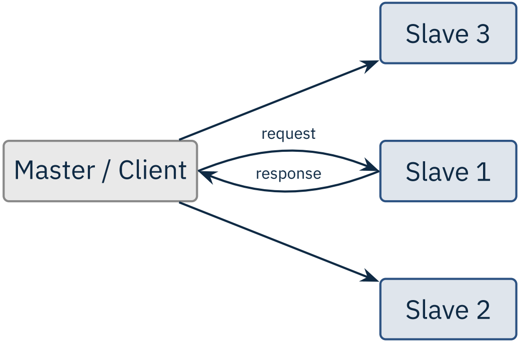
Modbus

Modbus Overview



Variants: Modbus RTU (serial), Modbus TCP (port 502)
Address space: coils, discrete inputs, input regs, holding regs
No auth, no encryption – by design (1979)

Source: Modbus Organization, *Modbus Application Protocol Specification* v1.1b3, 2012.

Modbus TCP Frame Anatomy

MBAP header (incl. Unit, 7 B)				PDU		
Trans ID	Proto ID	Length	Unit	Func	Addr	Count
`00 01`	`00 00`	`00 06`	`01`	`03`	`00 00`	`00 0A`

Decoded

"Read 10 holding registers starting at address `0x0000` from slave 1."

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Siemens, DNP3, EtherNet/IP

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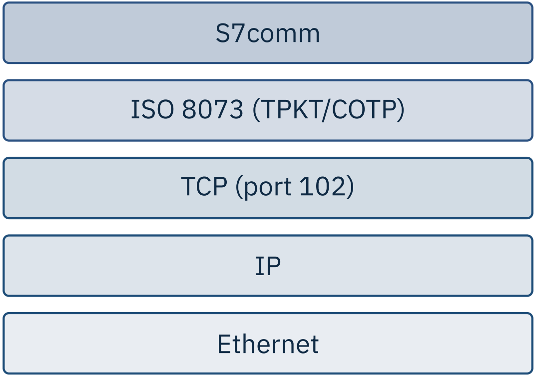
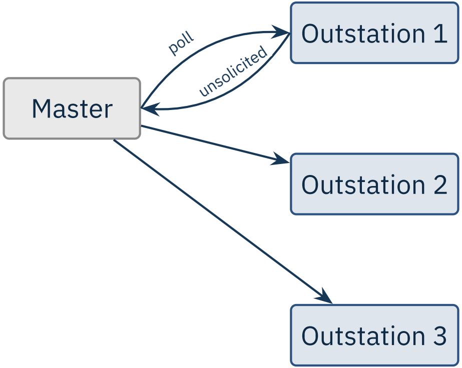
Siemens S7comm



Read/write variables, upload/download programs, start/stop CPU. S7commPlus added session keys – still broken in practice.

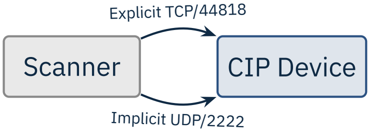
DNP3 and IEC 60870-5-104



DNP3 dominant in NA utilities; IEC 60870-5-104 in Europe. Both ride TCP today; both added Secure Authentication (SAv5) on top, but encryption is optional.

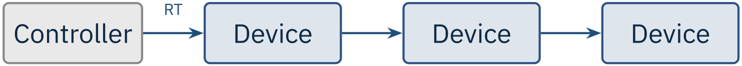
Source: IEEE 1815-2012 (DNP3) | IEC 60870-5-104:2006 + Amd. 1:2016.

EtherNet/IP and CIP



CIP carries object-oriented messages (assemblies, parameters, identity).
CIP Security adds TLS/DTLS and certificate-based authentication.

Profinet



EtherType 0x8892 – bypasses IP for real time.
IRT needs hardware switching for sub-ms cycles.

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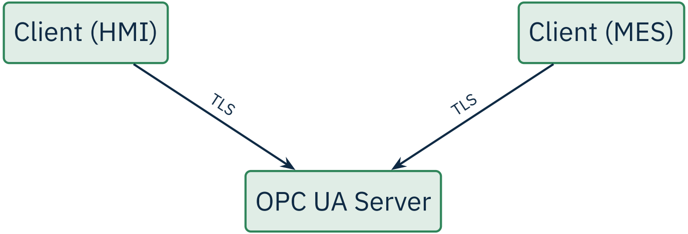
Modern Secure Protocols

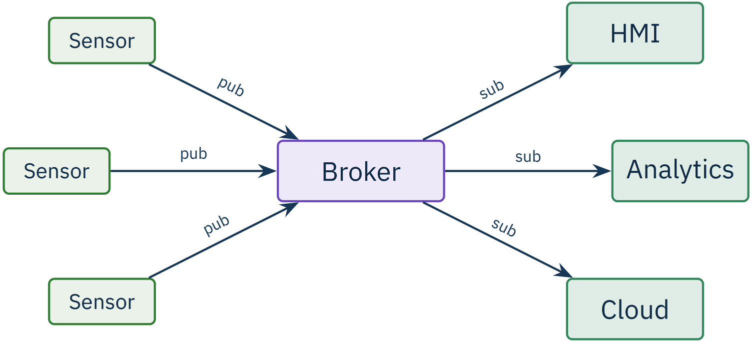
OPC UA



X.509 mutual auth • message signing • encryption
Rich information model • pub/sub since v1.04

Source: IEC 62541 (multipart, OPC Foundation) | OPC UA Specification Part 2 (Security Model).

MQTT for IIoT

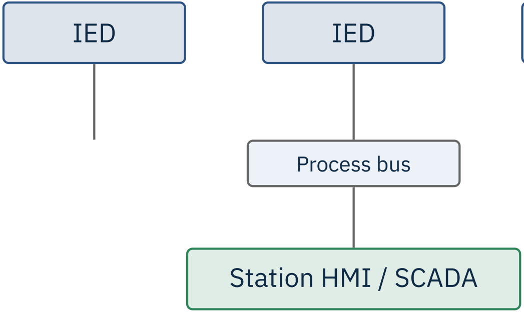


Default 1883 plaintext / 8883 with TLS. Sparkplug B adds OT semantics.

Protocol Security at a Glance

Protocol	Auth	Integrity	Confid.	Notes
Modbus	×	×	×	1979, ubiquitous
S7comm	×	×	×	Siemens-proprietary
DNP3	opt SAv5	opt	×	Utilities
EtherNet/IP	opt CIP	opt	opt	Rockwell
Profinet	weak	×	×	Siemens / IRT
OPC UA	X.509	sign	TLS	modern default
MQTT	+TLS	TLS	TLS	IIoT messaging

IEC 61850 in Substations



The substation stack

MMS for client/server; **GOOSE** for sub-4 ms trip messages; **Sampled Values** for measurements. Goose has *no* authentication by default.

Source: IEC 61850 (multipart) | IEC 62351 (security for IEC TC 57 protocols).

Summary – Section 04

★ Key Takeaway: What to remember from this section

- Most legacy industrial protocols are unauthenticated and unencrypted by design.
- Modbus is the canonical example – 12 bytes of MBAP and a one-byte function code.
- Modern protocols (OPC UA, MQTT-TLS) bring real cryptography, but require correct configuration.
- Network-layer mitigations (firewalls, segmentation) compensate where the protocol cannot.

References – Section 04

- Modbus Organization. *Modbus Application Protocol Specification* v1.1b3, 2012.
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End of Section 04

Industrial Protocols

Questions and discussion

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